Mon 31 May 2025

Last time we continued our discussion on finding fixed points of $g(x): \mathbb{R} \to \mathbb{R}$

We saw that if
$g \in C[a,b]$ and $g(x) \in [a,b]$ $\forall x \in [a,b]$,
then $g(x)$ has a fixed point in $[a,b]$
Furthermore, if $g'(x)$ exists on (a,b) and $\forall x \in [a,b]$
$|g'(x)| \le k < 1$, for some k, then the fixed point is unique.

We also saw that if $g(x^*) = x^*$, and
$|g'(x^*)| < 1$ x^* is an attracting fixed point (stable)
or
$|g'(x^*)| > 1$ x^* is a repelling fixed point (unstable)

$x_{i+1} = g(x^* + h) =$ then $|x_{i+1} - x^*| < h$
$x_{i+1} = g(x^* + h)$ then $|x_{i+1} - x^*| > h$.

We started to discuss two examples:

① $g_1(x) = \frac{1}{2}(x^3 + x)$ $g_1'(x) = \frac{1}{2}(3x^2+1)$
② $g_2(x) = -\frac{1}{2}(x^3 + x)$ $g_2'(x) = -\frac{1}{2}(3x^2+1)$

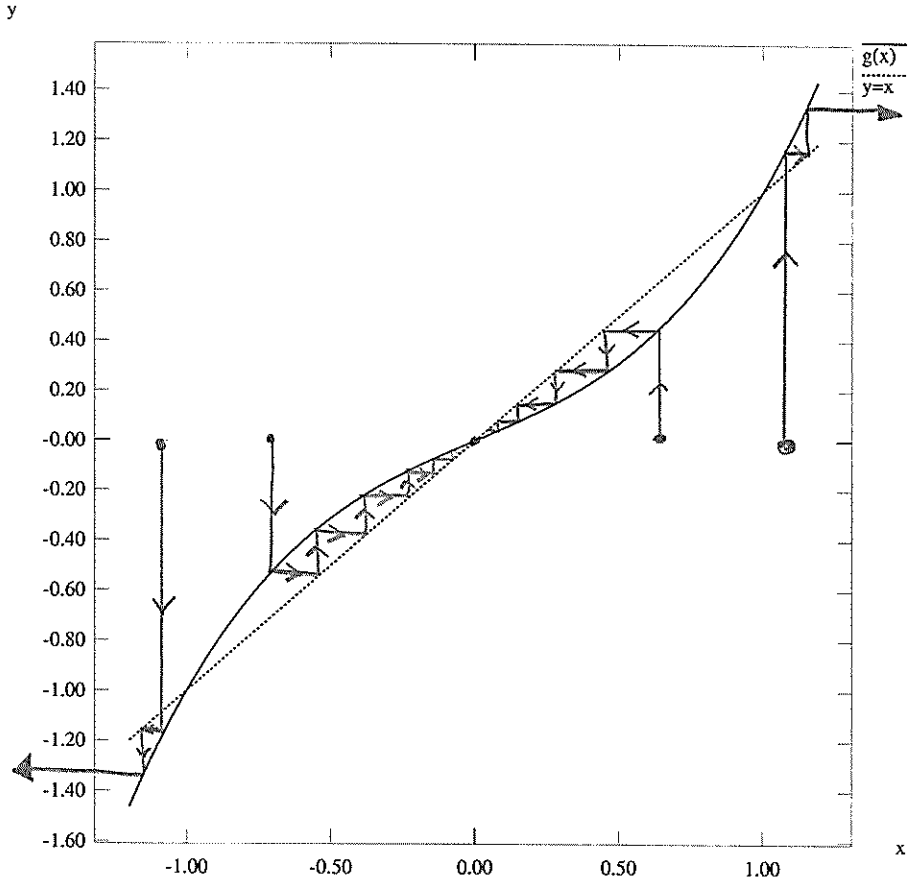
$g_1(x)$ has 3 fixed points: $g_1(0) = 0$ $g_1(1) = 1$ $g_1(-1) = -1$
$g_1'(0) = 1/2$ $g_1'(1) = 2$ $g_1'(-1) = 2$

The iteration $x_0 =$ given
$x_{i+1} = g(x_i)$
$\to -\infty$ for $x_0 < -1$
0 for $-1 < x_0 < 1$
$+\infty$ for $x_0 > 1$

$g_2(x)$ has 1 fixed point: $g_2(0) = 0$ $g_2'(0) = 1/2$
<u>note</u>: $g_2(-1) = +1$, $g_2(+1) = -1$ period 2 points
can draw functional iteration diagrams to confirm

sequence
$\to 0$ for $-1 < x_0 < 1$
$\to$ blows up for $x_0 < -1$, $x_0 > 1$

y = g(x) = 0.5 (x3 + x) and y = x**

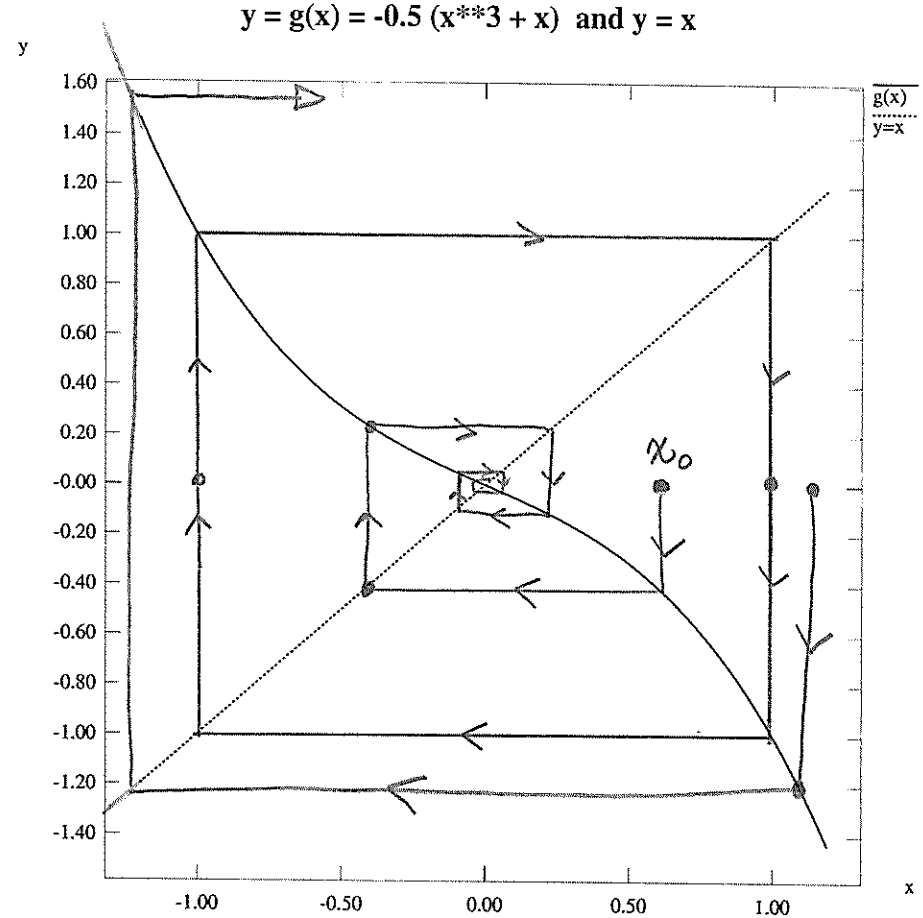


$g(0) = 0$ $\quad g'(0) = 0.5$

$g(1) = 1$ $\quad g'(1) = 2$

$g(-1) = -1$ $\quad g'(-1) = 2$

y = g(x) = -0.5 (x3 + x) and y = x**



$g(0) = 0$ $\quad g'(0) = -0.5$ $\quad g'(1) = -1$

$g'(-1) = -1$

$g(1) = -1$

$g(-1) = 1$ $\quad \pm 1$ are period-2 points

$g' = -\frac{1}{2}(3x^2+1)$

basin of attraction for $x^* = 0$ is $-1 < x_0 < 1$

We can summarize everything we've seen so far in the well-known:

<u>Fixed Point Theorem:</u>

Let $g \in C[a,b]$ be such that $g(x) \in [a,b]$ for all $x \in [a,b]$. Suppose, in addition, g is differentiable on (a,b) and $|g'(x)| \leq k < 1$, for some constant k and for all $x \in (a,b)$.

Then for any $x_0 \in [a,b]$, the fixed point iteration

$$x_{i+1} = g(x_i) \qquad i = 0, 1, 2, \ldots$$

converges to the fixed point $x^* \in [a,b]$

example: Apply the fixed point theorem to our familiar root finding problem $f(x) = x - 0.2\sin(x) - 0.5$
with $g(x) = 0.2\sin(x) + 0.5$ on the interval $[-1,1]$.

Theorem conditions:

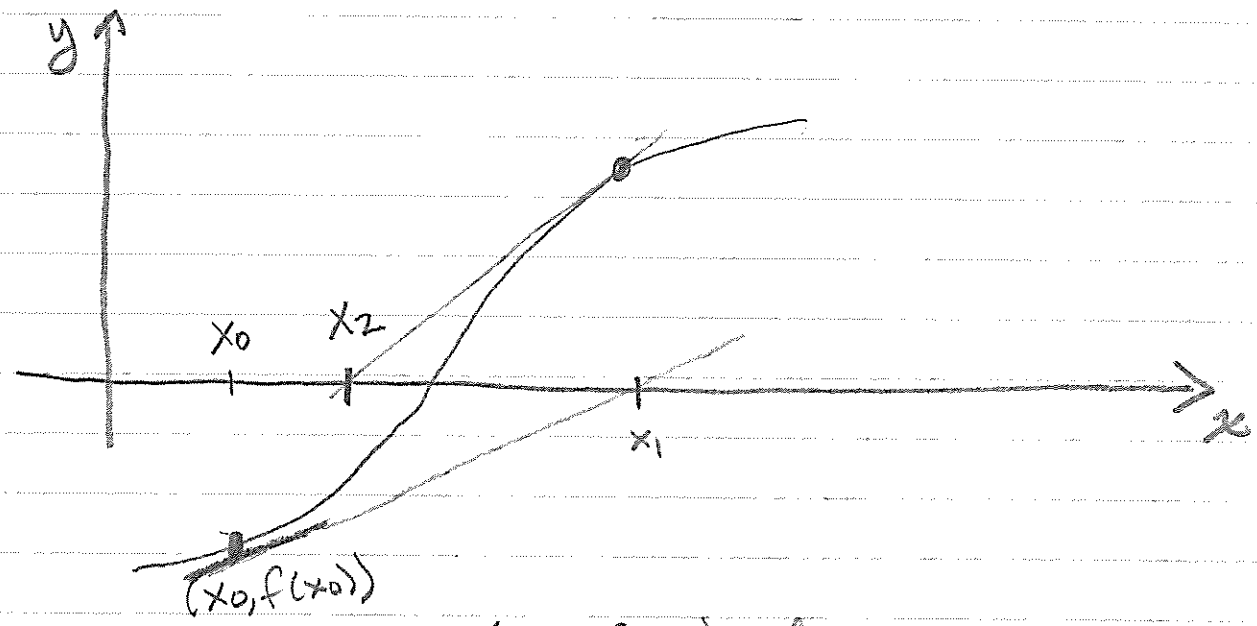
- $g(x)$ is continuous and differentiable on $[-1,1]$
- we showed earlier that $\forall x \in [-1,1]$, $g(x) \in [-1,1]$
- $g'(x) = 0.2\cos(x)$
- $\forall x \in [-1,1]$, $|g'(x)| = |0.2\cos(x)|$
  ≤ 0.2
  < 1

$\therefore$ by the fixed point theorem, the sequence
$x_{i+1} = g(x_i) \quad i = 0, 1, 2, \ldots$
converges for any $x_0 \in [-1,1]$

note: conditions are sufficient but not necessary.

Let's continue to apply this knowledge about fixed points to the root finding problem.

Suppose $f(x)$ has the following graph:



And suppose we know $(x_0, f(x_0))$ lies on the graph

$\boxed{\text{Let's try to approximate } f(x) \text{ near } x_0.}$

The straight line that goes through $(x_0, f(x_0))$ that has the same slope as $f(x)$ at x_0 has equation

$$y - f(x_0) = f'(x_0)(x - x_0)$$

Since this line approximates $f(x)$ near x_0, it is reasonable to suspect that its root may give a good approximation to the root of $f(x)$.

Setting $y = 0$ and solving for x gives

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Of course, x need not be a root of $f(x)$, so subscript it ($x \to x_1$) and take it as our next approximation to the root.

We can repeat this idea to get $x_2 = x_1 - \dfrac{f(x_1)}{f'(x_1)}$.

And then generate the iteration

$$\begin{cases} x_0 = \text{given initial guess} \\ x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}, \quad i = 0, 1, 2, \ldots \end{cases}$$

called Newton's method.

It is a fixed point iteration with $g(x) = x - \dfrac{f(x)}{f'(x)}$.

<u>example</u>: Apply Newton's method to

$$f(x) = x - 0.25\sin(x) - 0.5$$

with $x_0 = 0$ until $|x_{i+1} - x_i| < 5 \times 10^{-7}$.

[Recall: the bisection method with $[L_0, R_0] = [0, 1]$ needs 21 iterations to meet tolerance. Our functional iteration with $g(x) = 0.25\sin(x) + 0.5$ req'd 9 iterations

We have:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$= x_i - \frac{(x_i - 0.25\sin(x_i) - 0.5)}{(1 - 0.25\cos(x_i))}$$

Crunching out the iteration scheme gives:

i	x_i	$\Delta x_i = \lvert x_i - x_{i-1} \rvert$
0	0.0	
1	0.625	6.3×10^{-1}
2	0.6154745...	9.5×10^{-3}
3	0.6154682...	6.3×10^{-6} almost!
4	0.6154682...	2.8×10^{-12}

difference ≈ squaring.

note: • met the criteria for stopping in only 4 steps.
• the change in x_i decreases by a "square" factor each iteration.

This example shows why Newton's method is popular — it can converge very quickly.

Also: Compared to the bisection method (which only used sign(f(m)) information, Newton's method makes use of function value $f(x_i)$ and its derivative $f'(x_i)$

Finding derivatives is an error bound process, and is difficult when $f(x)$ is complicated. So let's keep looking for more methods.